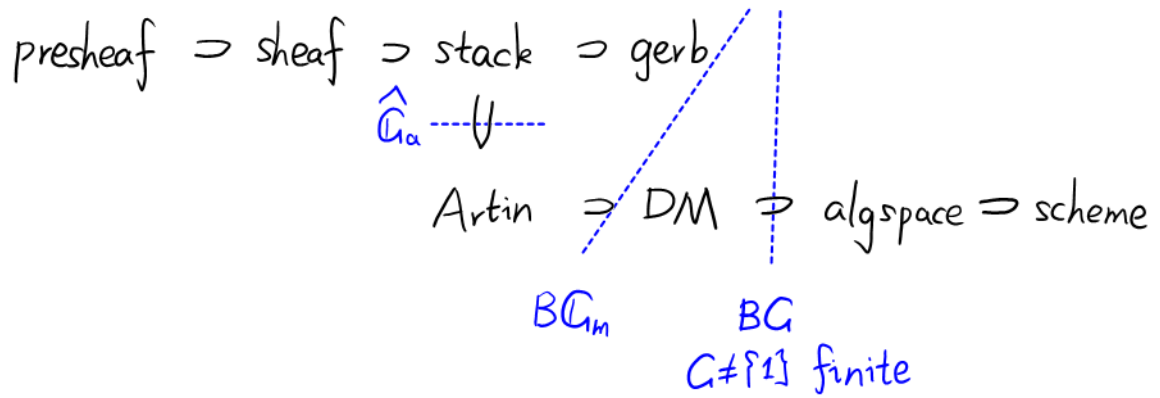


Eine Woche, ein Beispiel
 5.18 gerb & twisted sheaves

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$$\begin{aligned}
 \hat{\text{G}}_a &= \text{Spf } k[[t]] && \longrightarrow && \text{Spec } k[[t]] \\
 &= \varinjlim_n \text{Spec } k[t]/(t^n) && && \\
 &= \{ (t) \} && && = \{ (0), (t) \}
 \end{aligned}$$

Def. (stack in groupoids) Let $\mathcal{F} \in Sh(X)$.

$\mathcal{F}: Open(X)^{\mathcal{P}} \longrightarrow \mathcal{G}pd$ is a stack, if

$$\mathcal{F}(U) \xrightarrow{\sim} \operatorname{holim} \left(\prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_{ij}) \Rrightarrow \prod_{i,j,k} \mathcal{F}(U_{ijk}) \right)$$

as an equivalence of groupoids, where

$$\begin{aligned} Ob(\operatorname{holim}(\dots)) &= \{ \text{descent datum } (\mathcal{L}_i, \phi_{ij}) \} \\ &= \left\{ (\mathcal{L}_i, \phi_{ij})_{i,j} \mid \begin{array}{l} \mathcal{L}_i \in \mathcal{F}(U_i) \\ \phi_{ij}: \mathcal{L}_i|_{U_{ij}} \xrightarrow{\sim} \mathcal{L}_j|_{U_{ij}} \\ \phi_{jk} \circ \phi_{ij}|_{U_{ijk}} = \phi_{ik}|_{U_{ijk}} \end{array} \right\} \end{aligned}$$

$$\operatorname{Mor}((\mathcal{L}_i, \phi_{ij}), (\mathcal{L}'_i, \phi'_{ij})) = \left\{ (f_i)_i \mid \begin{array}{ccc} f_i: \mathcal{L}_i & \xrightarrow{\sim} & \mathcal{L}'_i \\ \mathcal{L}_i|_{U_{ij}} & \xrightarrow{f_i|_{U_{ij}}} & \mathcal{L}'_i|_{U_{ij}} \\ \phi_{ij} \downarrow & \circlearrowright & \downarrow \phi'_{ij} \\ \mathcal{L}_j|_{U_{ij}} & \xrightarrow{f_j|_{U_{ij}}} & \mathcal{L}'_j|_{U_{ij}} \end{array} \right\}$$

E.g.

$$Sh^{\wedge} = QCoh$$